

Red giant

Different stars change in different ways. How a star changes depends on how much gas it is made of. When stars like the Sun have used up all their hydrogen, their surface will cool and turn red. They will swell up and become red giants. Don't worry though, this won't happen to the Sun for another five billion years!

White dwarf →

White dwarf

The red giant Sun will eventually use up all of its fuel. Then it will shrink and become a white dwarf. All the material will pack so close together that the dying Sun will be smaller than Earth.

Supernova

Some stars more massive than the Sun end their lives by blowing themselves apart. When a star explodes like this, it is called a supernova.

The material pushed away in the explosion will produce new stars. →

Beware of the black hole!

As some stars die, their material gets more and more squashed together. The stars shrink until they become a point in space. This is called a black hole. Its gravity is so strong that anything that gets too close is sucked into it and can never escape.

